

Coastal Hydrodynamics V

Course : CE60222

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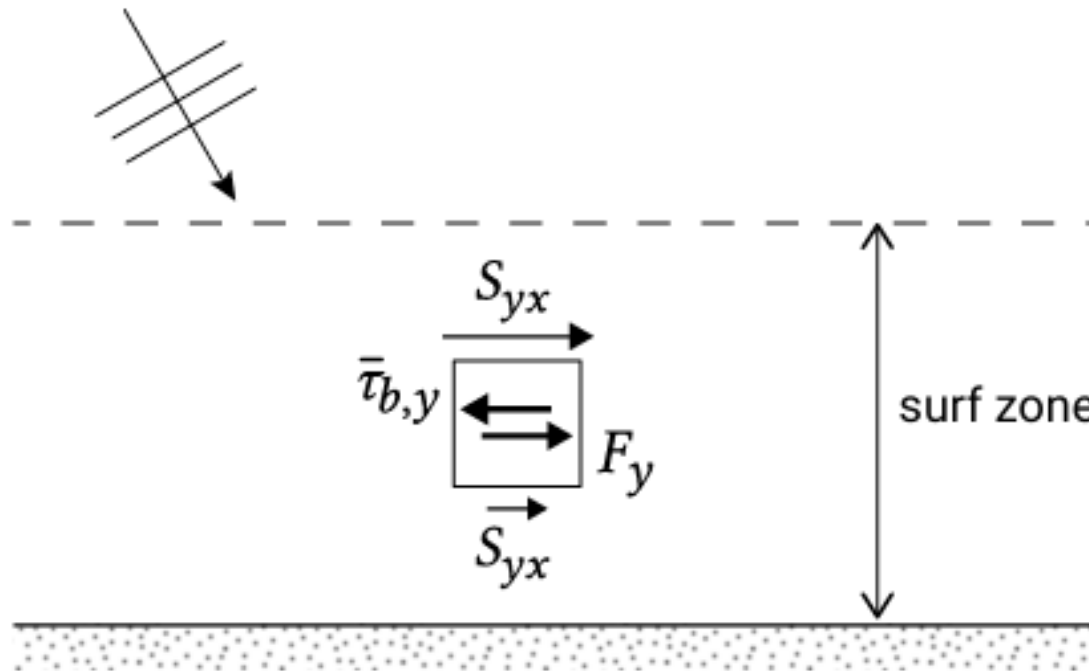
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Alongshore balance: longshore current

- In the **alongshore direction**, the **transfer of momentum** from the wave motion to the mean flow gives rise to a **longshore current**.
- The **longshore current** (velocity magnitude and cross-shore distribution) is an **important input parameter** in **longshore sediment transport** computations.

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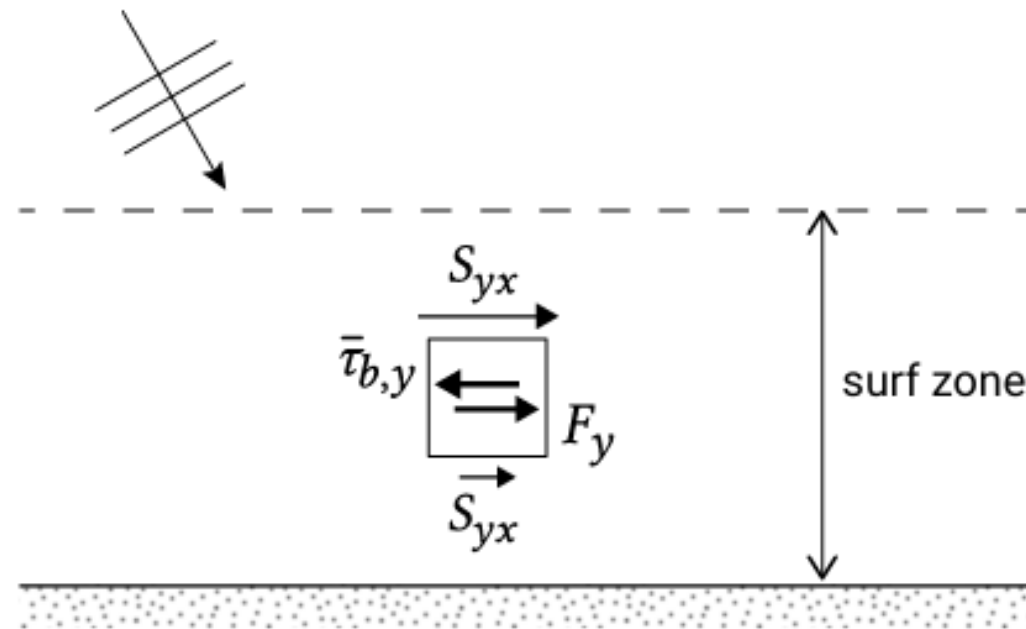
Alongshore momentum balance (straight, parallel depth contours)

- In a 2D scenario with **long-crested** waves **obliquely incident** to an **alongshore uniform coast**, the cross-shore variation of the **radiation stress shear** component S_{yx} drives a **longshore current**.
- In the **cross-shore direction**, this force is balanced by a **hydraulic pressure gradient**.
- Along an **infinitely long coastline**, no alongshore pressure gradient can form, so the **balancing force** is provided by **bed shear stresses**, which arise only when a **longshore current is generated**.
- These **shear stresses restrain the current**, while in the **cross-shore direction**, bed shear stress is **negligible** compared to the pressure force.

- The **alongshore component** of the momentum balance for a **steady state** and alongshore uniformity can be written as:

$$F_y = -\frac{dS_{yx}}{dx} = \bar{\tau}_{b,y}$$

- The balance between the **driving force** and the **resisting or retarding force** is shown in Figure.



Wave force in alongshore uniform situation

- Let us first assess $F_y = -\frac{dS_{yx}}{dx}$ in **deeper water**, i.e. outside the **breaker zone**.

where,

$$S_{yx} = En \sin \varphi \cos \varphi$$

and φ = the wave angle changes as a **function of x** , but does **not vary with y** .

- The latter leads to **Snell's law** for **regular waves** :

$$\sin \varphi / c = \text{constant}$$

- Conservation of energy requires under the present assumptions.

$$\frac{d}{dx} (Ec_g \cos \varphi) = -D_w$$

- For **linear waves** we can now write the **wave force** as:

$$F_y = -\frac{dS_{yx}}{dx} = -\frac{\sin \varphi}{c} \frac{d}{dx} E c_g \cos \varphi = \frac{D_w}{c_0} \sin \varphi_0$$

- Apparently the **alongshore driving force** is a function of the **dissipation** of the **wave energy**.
- **Outside the surf zone**, the **dissipation** of wave energy can be **neglected** and hence the **energy flux** is constant.

$$E c_g = (E c_g \cos \varphi, E c_g \sin \varphi) = \text{Constant}$$

- In the **absence of dissipation**:

S_{yx} is constant

$$F_y = 0.$$

- So, although **outside the surf zone** the wave conditions **change with x** (wave height due to shoaling; wave direction due to refraction), the **radiation shear stress** is constant.
- Therefore, since the **alongshore forcing** is only present when the waves are breaking, the **longshore current** is confined to the **surf zone**.

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Analytical model for alongshore wave force in the surf zone

- To simplify the alongshore wave force, a **basic model for wave breaking** is used, where **wave height** H is proportional to **water depth** h in the surf zone.

$$H = \gamma h$$

- An alternative would be solving the **energy balance** numerically. **Snell's law** and the simple dissipation model result in:

$$F_y = -\frac{\sin \varphi}{c} \frac{d}{dx} E c_g \cos \varphi$$

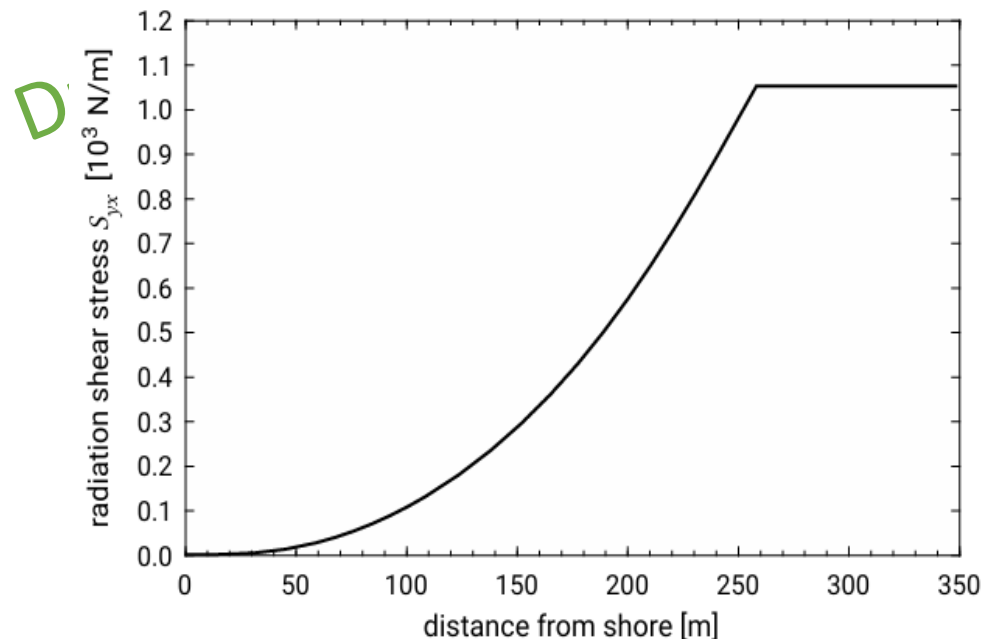
$$F_y = -\frac{\sin \varphi_0}{c_0} \frac{d}{dx} \frac{1}{8} \rho g \gamma^2 h^2 c_g \cos \varphi$$

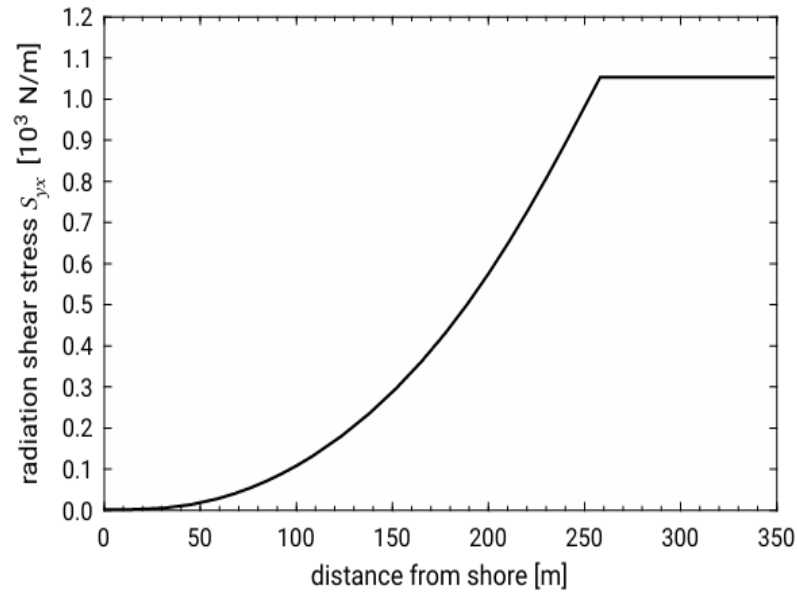
- In shallow water $c_g = c = \sqrt{gh}$ and since, due to refraction, φ is small (generally around 10° to 15°), we assume $\cos \varphi \approx 1$. We now have:

$$F_y \approx -\frac{\sin \varphi_0}{c_0} \frac{1}{8} \rho g^{3/2} \gamma^2 \frac{d}{dx} h^{5/2}$$

$$F_y = -\frac{5}{16} \frac{\sin \varphi_0}{c_0} \rho (gh)^{3/2} \gamma^2 \frac{dh}{dx}$$

- The **radiation shear stress** S_{yx} is constant seaward of the border of the breaker zone (thus the wave force is zero). It **decreases** inside the breaker zone to **zero** at the waterline.





- Radiation shear stress S_{yx} for $H_0 = 2\text{m}$, $T = 7\text{s}$, $\varphi_0 = 30^\circ$, constant bottom slope 1:100 and a breaker index $\gamma = 0.8$.
- For many day to day wave conditions, this gradient in S_{yx} causes alongshore stresses (forces) which are in the same order of magnitude as the bottom shear stress in rivers: in the order of 1N/m^2 to 10N/m^2 .
- The cross-shore gradient in the alongshore radiation stress S_{yx} is therefore an important driving force in the littoral zone.

Quadratic friction law for bed shear stress

- To determining the alongshore current, we need to calculate the **bed shear stress**.
- **Turbulence** from **currents** spreads through the **whole water**, while **wave** turbulence stays **near the seabed** and makes the bed shear stress much stronger.
- This happens because waves **move faster** and create **more friction**.
- To describe the **longshore current's** vertical profile, we use a **quadratic law** that combines the effects of **both currents and waves**.

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Quadratic friction law for bed shear stress is non-trivial since:

- The **quadratic friction law** makes the **combined effect of waves and currents non-linear**, often resulting in a total shear stress greater than the sum of their individual stresses.
- Waves and currents can move in **different directions**, with waves approaching at an angle and currents flowing along the coast, causing the **bed-shear stress to vary based on their angle**.
- Currents create **turbulence throughout the water**, while waves **only affect the wave boundary layer**, raising the question of which velocity to use for combined motion.
- Most models focus on **time-averaged bed shear stress**, ignoring short-term changes within a wave cycle. However, these intra-wave changes can **impact sediment transport**.
- **Different models describe bed shear stress** in various ways—some focus **on time-averaged stress**, others **on instantaneous stress**.
- Accurate **bed shear stress** calculations are crucial for **predicting sediment transport**.

For the determination of the time-averaged bed shear stress in the alongshore direction, necessary to compute the longshore current, we take a fairly simple approach:

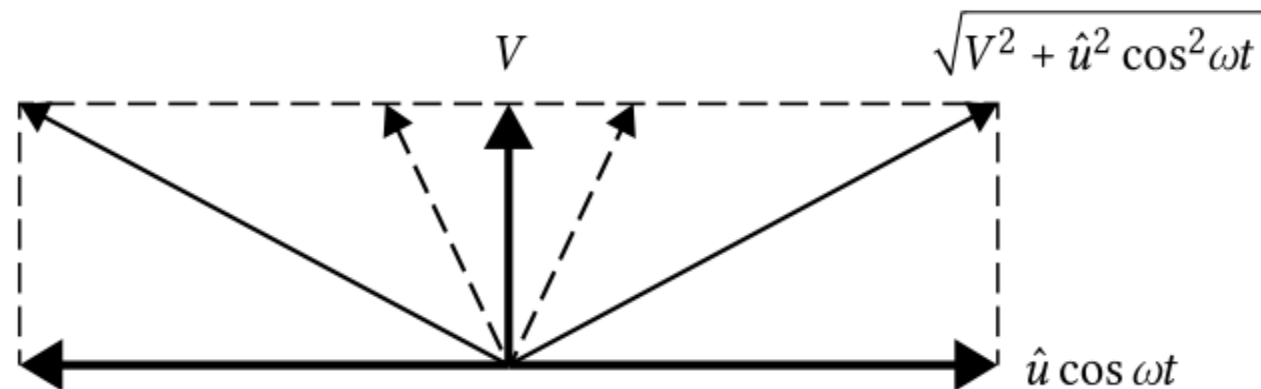
- The **wave motion** is described by **shallow-water theory** (constant orbital amplitude outside the wave boundary layer);

- The **angle of incidence** is **very small**, such that for the wave motion

$$(u_x, u_y) = (\hat{u} \cos \omega t, 0)$$

- The **bed friction vector** is related to the **depth-averaged velocity vector**. The latter is the sum of the depth-averaged longshore current velocity and the wave orbital motion:

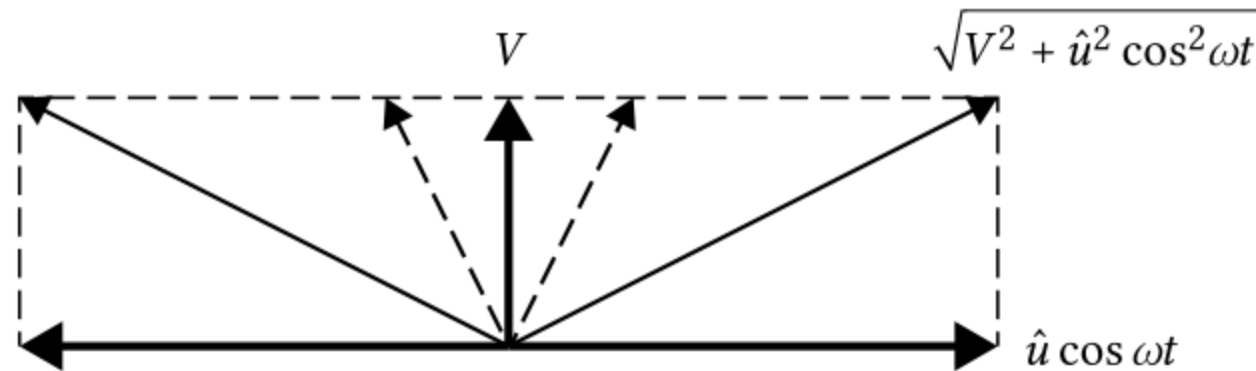
$$\vec{u} = (\hat{u} \cos \omega t, V)$$



- The **time-varying bed friction** is written as:

$$\vec{\tau}_b = \rho c_f |\vec{u}| \vec{u}$$

- The enhancement of the **friction factor** (compared to a current-only situation) due to the **small height** of the **wave boundary layer** as compared to the **current boundary layer** is not further specified.



- In the **cross-shore direction**, the time-averaged (not the instantaneous) bed shear stress is zero.
- With the above approximations, the time-averaged bed shear stress in the alongshore direction reads:

$$\bar{\tau}_{b,y} = \overline{\rho c_f |\vec{u}| V} = \overline{\rho c_f \sqrt{V^2 + \hat{u}^2 \cos^2 \omega t} V}$$

- If we further assume that $V \ll \hat{u}$, this can be simplified to:

$$\bar{\tau}_{b,y} = \frac{2}{\pi} \rho c_f \hat{u} V$$

- With $\hat{u}$ in shallow water and with a constant ratio of wave height over water depth across the entire surf zone we find:

$$\bar{\tau}_{b,y} = \frac{1}{\pi} \rho c_f \sqrt{gh} \frac{H}{h} V$$

Analytical model for longshore current (no lateral dispersion)

- For steady conditions, the alongshore velocity follows from the balance between the driving force and the resisting friction force.

$$F_y = -\frac{dS_{yx}}{dx} = \bar{\tau}_{b,y}$$

- This yields with Equations

$$F_y = \frac{D_w}{c_0} \sin \varphi_0$$
$$\bar{\tau}_{b,y} = \frac{1}{\pi} \rho c_f \sqrt{gh} \frac{H}{h} V$$

Now ,

$$\frac{D_w}{c_0} \sin \varphi_0 = \frac{1}{\pi} \rho c_f \sqrt{gh} \frac{H}{h} V$$

$$V(x) = \frac{\pi}{c_f \rho \sqrt{g}} \frac{\sin \varphi_0}{c_0} \frac{D_w(x)}{H(x)} \sqrt{h(x)}$$

- The magnitude of the depth-averaged longshore current velocity varies in the surf zone as a function of the dissipation, wave height and water depth.
- The dissipation and wave heights can be modelled using a wave model (with roller model).
- In our simple dissipation model

$$\gamma = H / h = \text{constant}$$

- we can write the force balance as:

$$-\frac{5}{16} \frac{\sin \varphi_0}{c_0} \rho (gh)^{3/2} \gamma^2 \frac{dh}{dx} = \frac{1}{\pi} \rho c_f \sqrt{gh} \gamma V$$

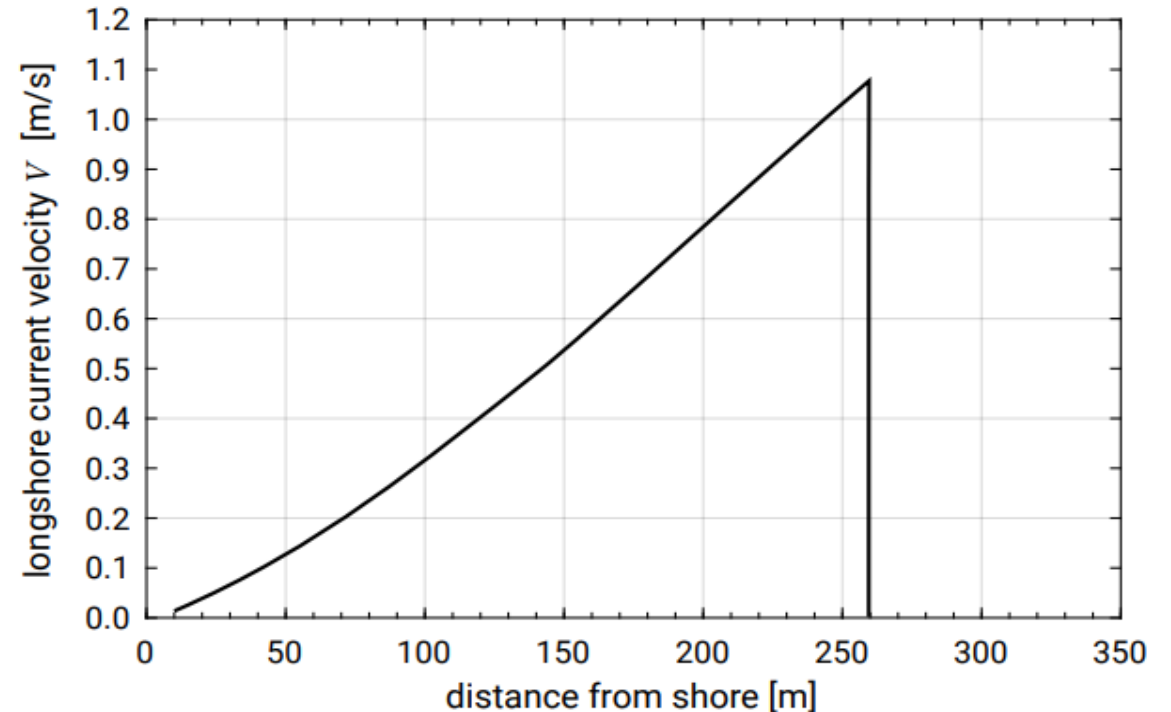
which leads to:

$$V(x) = -\frac{5}{16} \pi \frac{\gamma}{c_f} g \frac{\sin \varphi_0}{c_0} h \frac{dh}{dx}$$

- For a constant beach slope $\tan \alpha = -dh_0/dx$ and for $dh/dx \approx dh_0/dx$, the current velocity is proportional to the depth with a maximum at the breaker line (where $h = h_b$):

$$V(x) = \frac{5}{16} \pi \frac{H_b}{c_f} g \frac{\sin \varphi_0}{c_0} \frac{h}{h_b} \tan \alpha$$

- A **longshore current profile** according to **above equation** is shown in **Figure**.
- Alongshore velocity distribution** (regular wave field, $H_0 = 2$ m, $T = 7$ s, $\varphi_0 = 30^\circ$, bottom slope 1:100, $\gamma = 0.8$, roughness height $r = 0.06$ m).



- Larger **wave heights at breaking** (H_b) increase the maximum longshore current velocity and widen the **littoral zone**, leading to greater water discharge in the surf zone.
- **The beach slope ($\tan\alpha$)** also affects the current velocity $V(x)$: steeper slopes increase velocities but narrow the surf zone, keeping the **total discharge** roughly **constant**.
- For **small wave angles**, the **longshore current velocity** at a given depth becomes a **linear function** of the wave angle φ_0 .

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Turbulent forces redistributing momentum

- Turbulence plays a significant role in fluid dynamics by **smoothing out velocity gradients** through **lateral dispersion of momentum**.
- When **modeling turbulence**, the total velocity is often divided into **mean, wave, and turbulent components**.
- Turbulent shear stress arises from **averaging over turbulent motions** and is modeled using an **eddy viscosity (ν_T)**, which is analogous to **molecular viscosity (ν)** but represents turbulent friction.
 - **Molecular Viscosity (ν)**: Measures the fluid's resistance to flow due to internal friction.
 - **Eddy Viscosity (ν_T)**: Represents turbulent friction and is much larger than ν in coastal waters ($\nu_T \gg \nu$).

- The eddy viscosity depends on:
 - **Characteristic Velocity**: Often related to wave orbital motion in coastal zones.
 - **Characteristic Length Scale**:
 - For **vertical mixing**, it is the water depth.
 - For **horizontal mixing**, it is not limited by depth, so **horizontal eddy viscosity** (ν_T, H) is typically much larger than **vertical eddy viscosity** (ν_T, V).
- The shear component of the radiation stress S_{yx} was defined through:

$$S_{yx} = \overline{\int_{-h_0}^{\eta} (\rho u_y) u_x dz}$$

where the velocity components are due to the orbital motion.

- In analogy with above equation, we can write for the turbulent force:

$$S'_{yx} = \overline{\int_{-k_0}^{\eta} (\rho u'_y u'_x) dz}$$

where the overbar now represents averaging over the turbulent motion (indicated with primes).

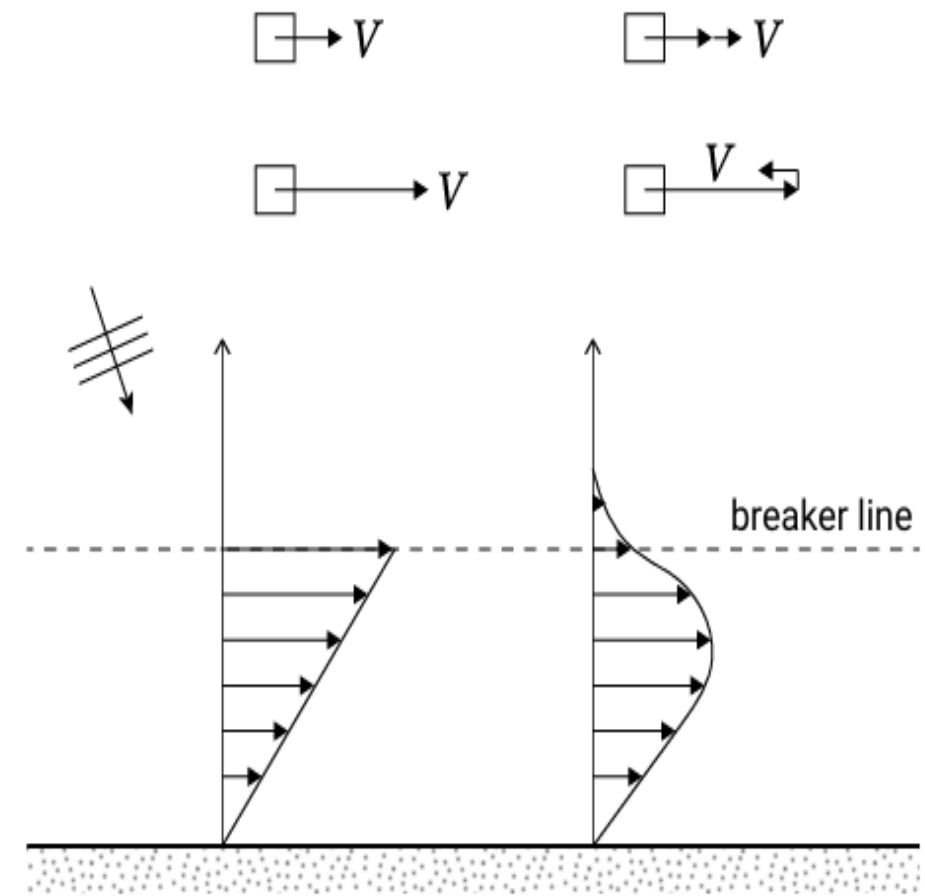
- This shear stress or friction force per unit surface area, acts on a surface parallel to the coast. It can be modelled as:

$$S'_{yx} \cong h \rho v_{T,H} \frac{dV}{dx}$$

- The eddy viscosity $\nu_{T,H}$ [m²/s] is also referred to as horizontal diffusivity.
- The momentum equation in the alongshore direction now reads:

$$\frac{D_w}{c_0} \sin \varphi_0 + \frac{d}{dx} \left(h \rho v_{T,H} \frac{dV}{dx} \right) = \bar{\tau}_{b,y}$$

- The **effect of turbulent forces**, smoothing the **longshore current profile**, is indicated in Figure.
- The **largest velocity gradient** at the **breaker line** causes the maximum transfer of **horizontal momentum**, reducing the **peak velocity** and shifting it landward.
- This also creates **longshore currents** outside the **breaker zone**.
- The **cross-shore distribution of eddy viscosity** (ν_T) further **influences** the velocity profile.

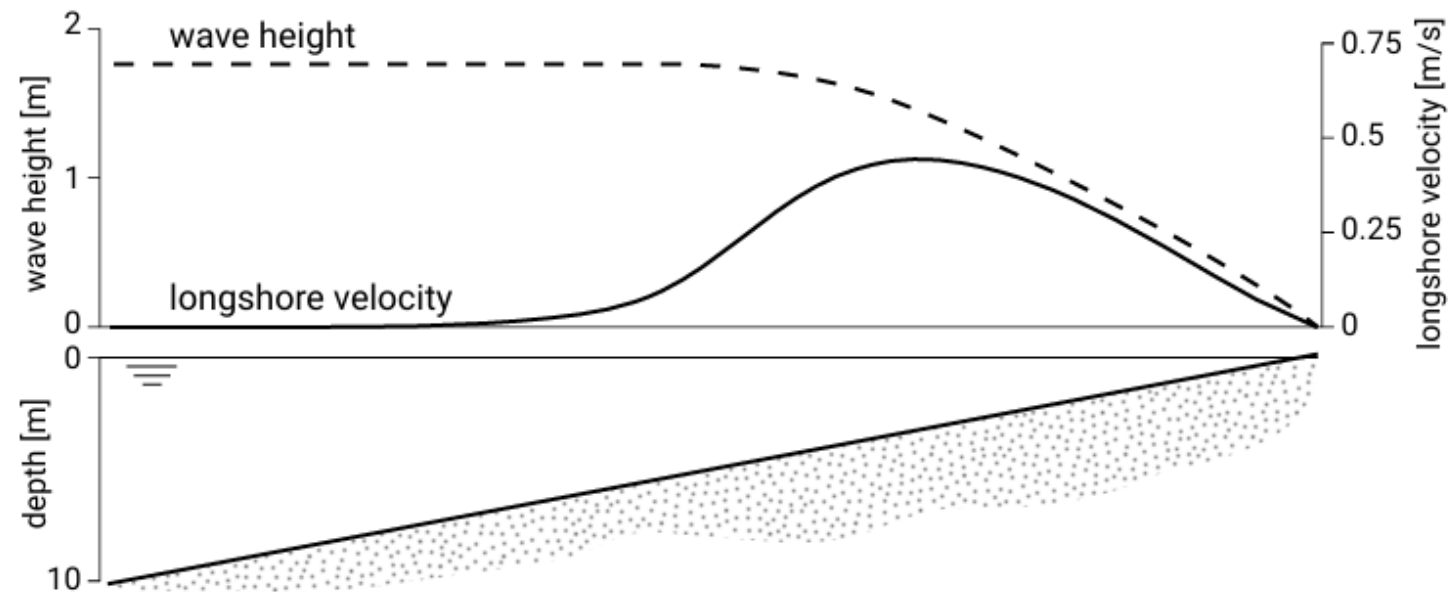


Roller momentum

- **Wave set-up** is observed to **start closer to the coast** than expected.
- This **spatial lag** was attributed to the **roller momentum**, which had **not been** taken into account.
- Similarly **longshore current velocity profiles** show an **onshore shift** in the maximum longshore current velocity.
- This can be **modelled** by including the **roller contribution** in the **alongshore momentum equation**.

Irregular waves

- Until now we have **only considered regular waves**. In reality, of course, **waves are irregular** and there is **no sharply defined breaker line**.
- **The effect of wave irregularity** is therefore to **smooth out the velocity distribution**, very similar to the **effect of turbulence**, giving a **wider and less sharply peaked** velocity distribution.
- This is also illustrated in **Figure**, which shows the output of a computation with the **computer model Unibest-CL+**.

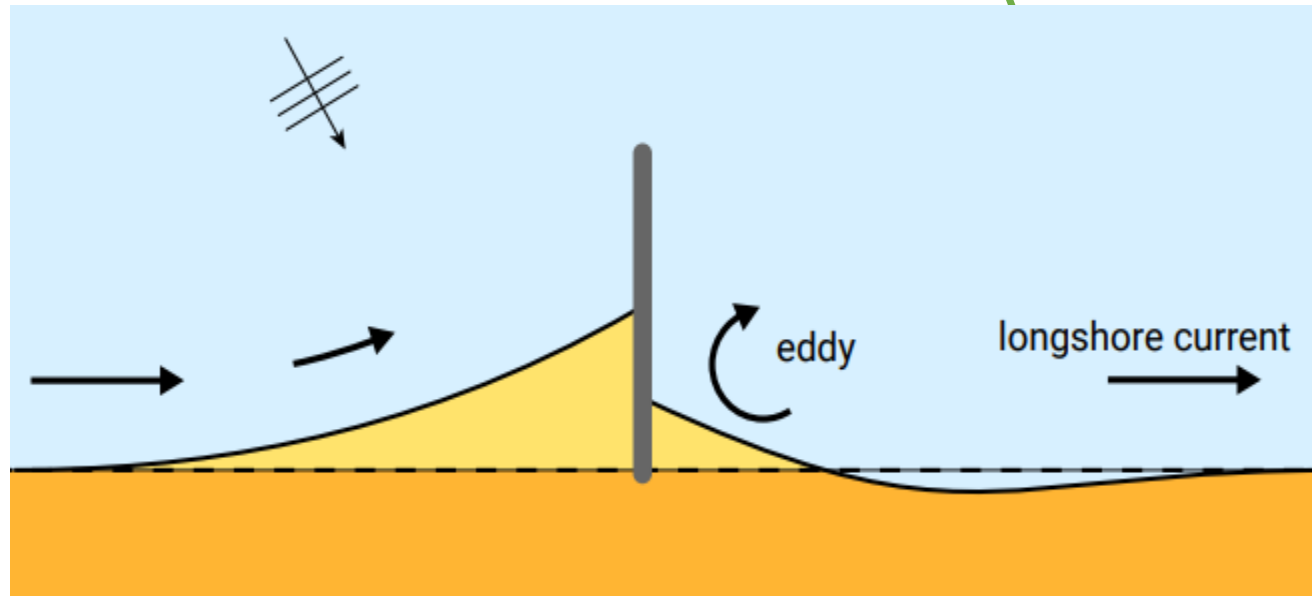


3D effects

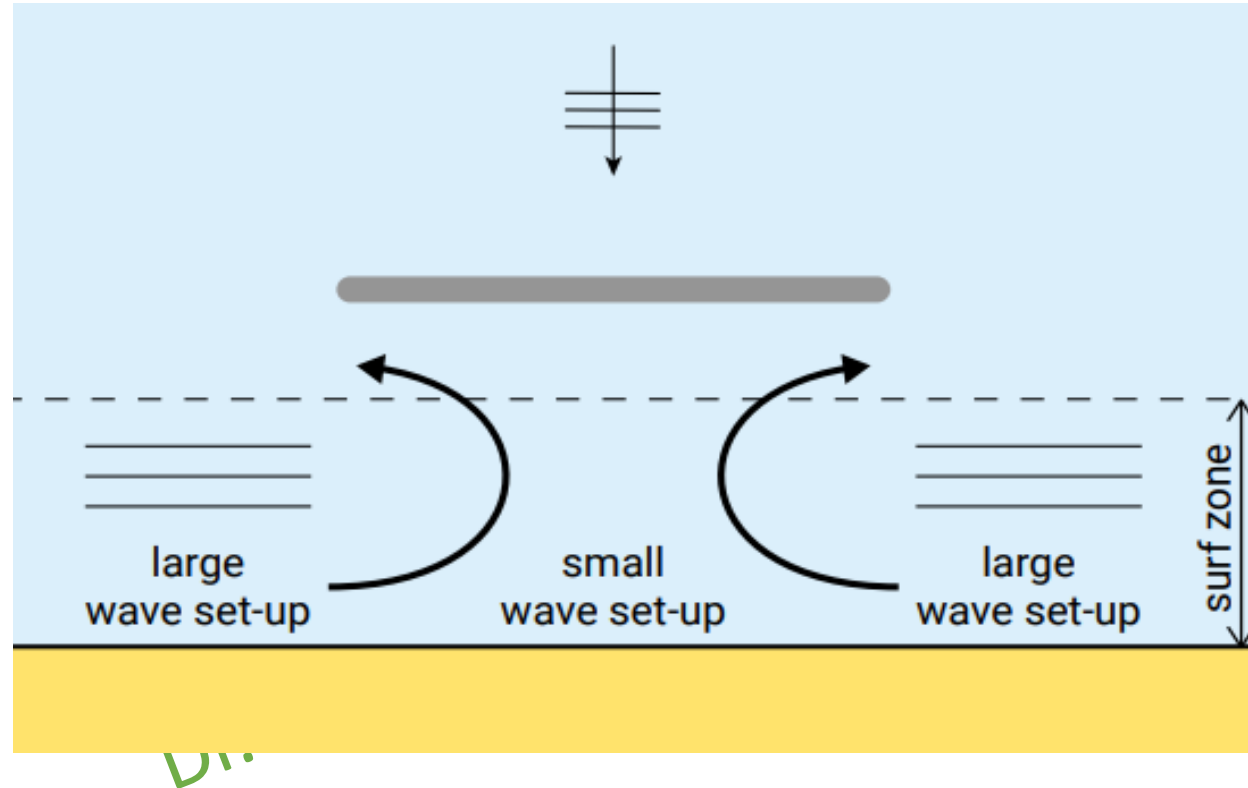
- Wave conditions can vary along the coast due to factors like **wave refraction** over uneven sea beds or **wave diffraction** near structures. These changes affect **wave-induced forces**, making terms like $\partial S_{yy}/\partial y$ and $\partial S_{xy}/\partial y$ may be non-zero.
- Variations in **wave height** along the coastline also cause differences in **cross-shore wave forces** and **wave set-ups**.
- Consequently, **pressure gradients** $\rho g h \partial \eta / \partial y$ occur along the coast and 3D current patterns are the result. We will briefly discuss the following situations:
 - i) **Eddy formation** in the shadow zone of structures;
 - ii) Creation of **rip currents**;
 - iii) **3D current patterns** around shoals.

Eddy formation in the shadow zone of structures

- In the case of **wave sheltering** due to for instance **groynes or detached breakwaters**, wave set-up can be expected to be **less in the sheltered area** than in the unsheltered region.
- This generates **local nearshore currents** towards the sheltered area.



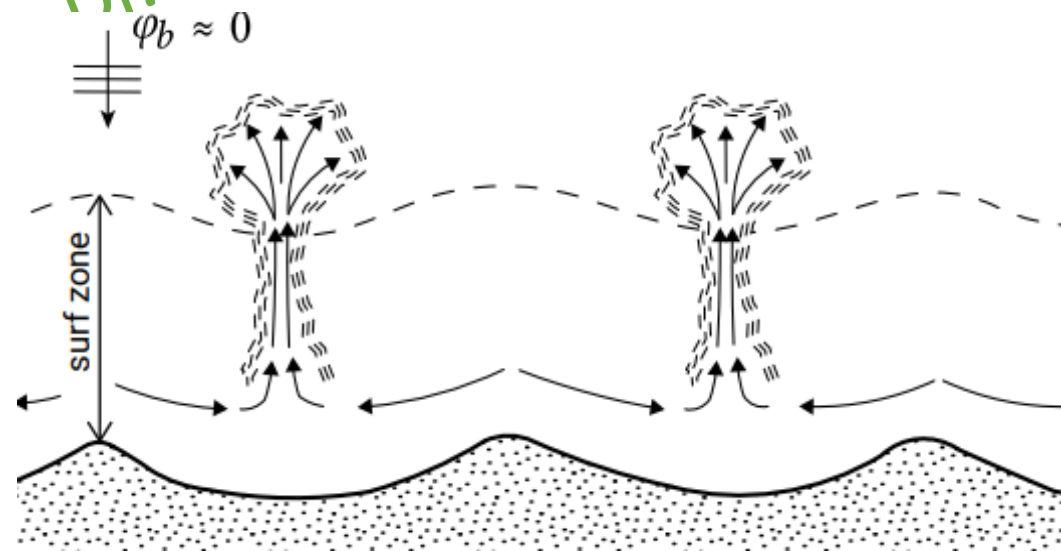
- In the **shadow zone of the groyne**, due to set-up differences a current runs towards the groyne, until it is diverted outward along the structure, **creating an eddy**.



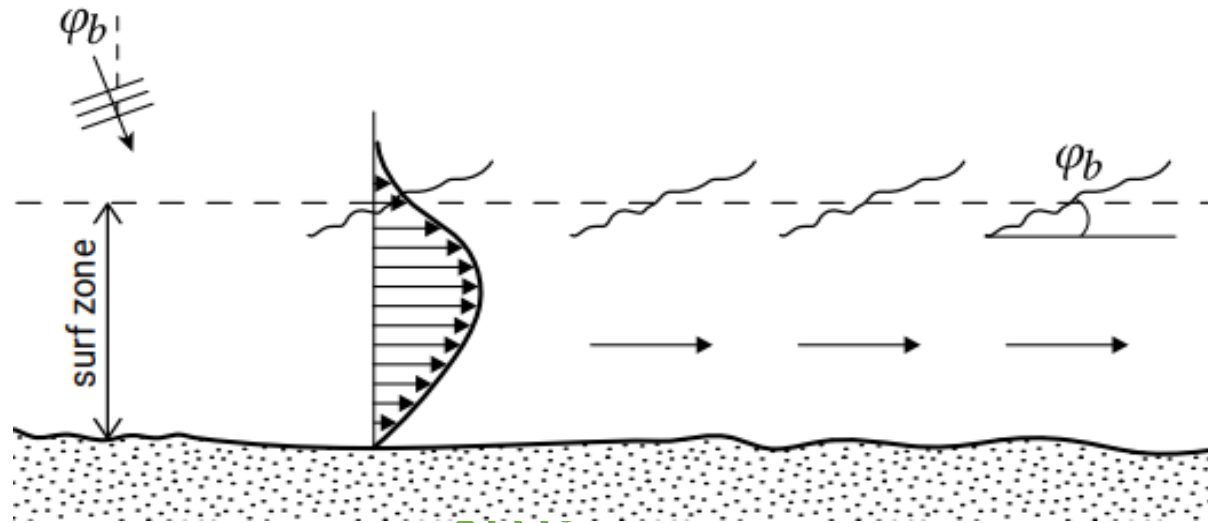
- In Figure **set-up differences** create **nearshore currents** towards the sheltered zone from **both sides** of the detached (emerged) breakwater.
- Therefore, a **return flow** will be present in deeper water, resulting in the development of two eddies.

Rip currents due to convergence or divergence of wave energy

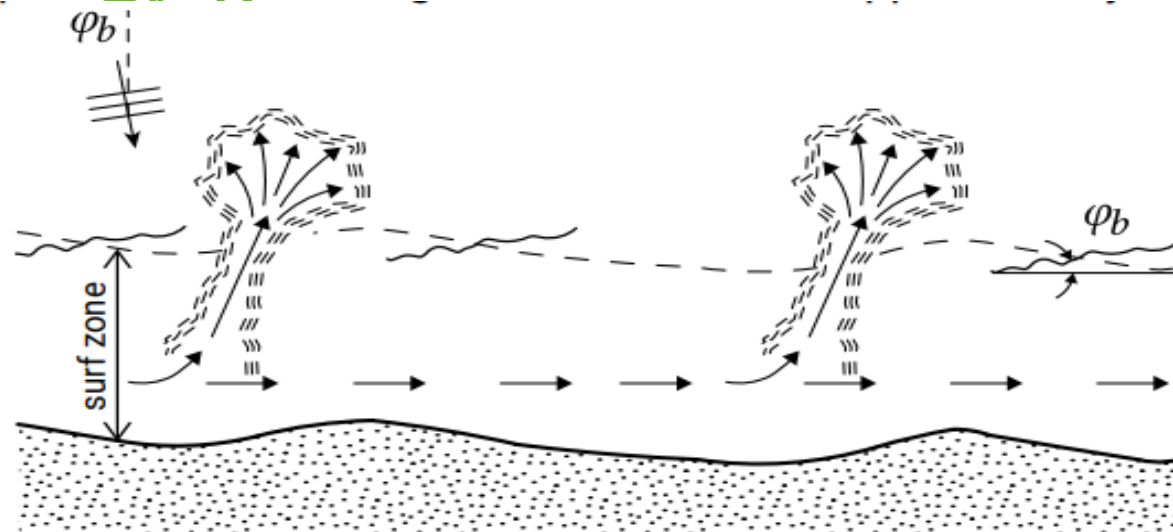
- **Rip currents** are strong, narrow flows moving seaward from the surf zone, **fed by longshore currents** that turn offshore.
- **Longshore currents** form due to **wave set-up differences**, flowing from high to low set-up areas.
- **Variations in set-up arise** from wave energy convergence/divergence due to **depth refraction or sheltering** (e.g., headlands).
- **An undulating coastline** focuses wave energy at **peaks**, creating rhythmic rip currents. This pattern develops mainly under near-normal wave incidence.



- In the case of **obliquely incident waves**, a longshore current driven by gradients in radiation shear stress (see Figure) will overshadow the more subtle effects of set-up differences.

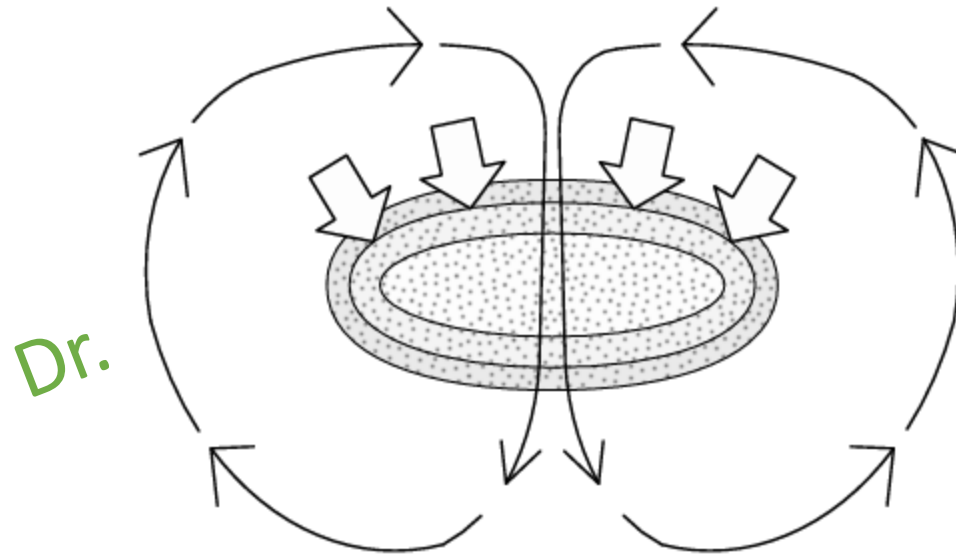


- A **combination of the two effects** may occur for slightly oblique wave incidence.

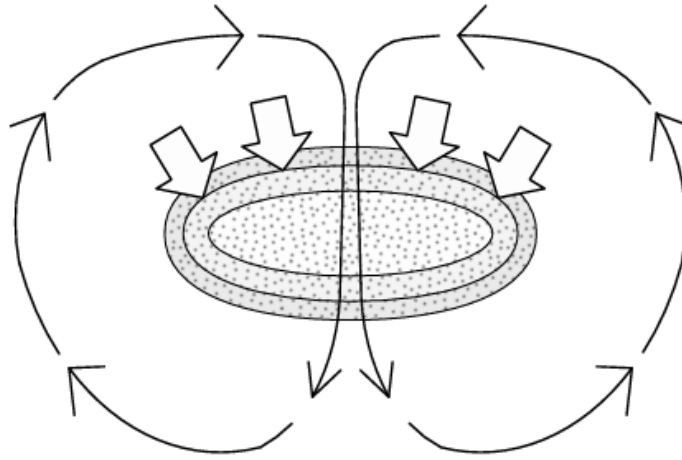


Wave-induced currents around shoals (and submerged breakwaters)

- In case of a **complex topography** with interrupted breaking, the pattern of **wave-induced currents** is **more complicated**.
- A still relatively **simple example** concerns a **shoal** on which waves are **breaking** (see figure).

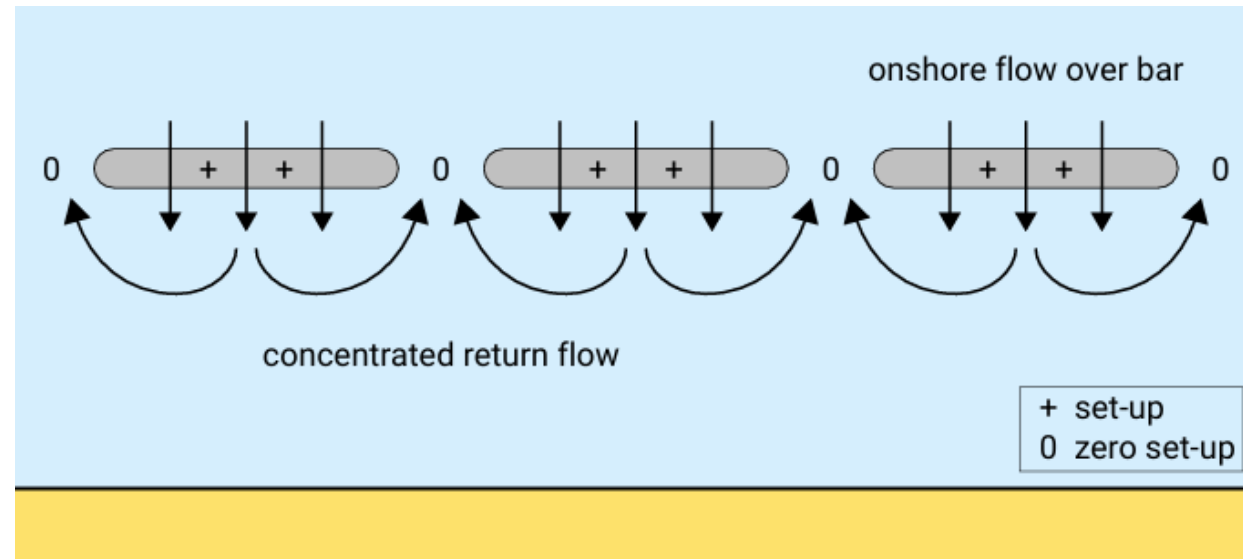


- Due to **refraction**, the waves will tend to **converge** toward the **top of the shoal**.
- As the **waves break** on the seaward slope of the shoal, they generate a **dissipation-related wave force**.



- At the **top of the shoal**, there is **no closed boundary** that requires a zero mean flow.
- Instead, the water will **flow over the shoal** in the direction of the force.
- The water **flows over the shoal** until it **reaches the channel behind the shoal**, where water level gradients will deflect it and drive it to **the sides of the shoal**.
- There it has room to **flow seawards again**, thus **closing the circulation**.
- In tidal inlets systems the wave-driven currents around shoals on the outer delta can be so strong that they dominate the tidal residual currents.

- In a similar fashion, the **interruption of wave-breaking** can generate **rip currents** along a stretch of coast.
- **Non-homogeneous wave-breaking** can occur in the case of a non homogeneous alongshore bar system or a (series of) submerged breakwater(s) on which waves break, see Figure.



- **Wave-breaking** on the bar induces a **set-up** over the bar as well as an onshore flow.
- Water level gradients and continuity force the flow to deflect to the sides and return seaward in between the breakwaters or bars in a concentrated rip current.

Wind-induced set-up and currents

- Moving **air exerts a shear stress τ** on the water surface that can be modelled by, again, a quadratic friction law:

$$\tau_{\text{wind}} = C_d \rho_a W^2$$

Where:

τ_{wind} = wind shear stress in N/m²

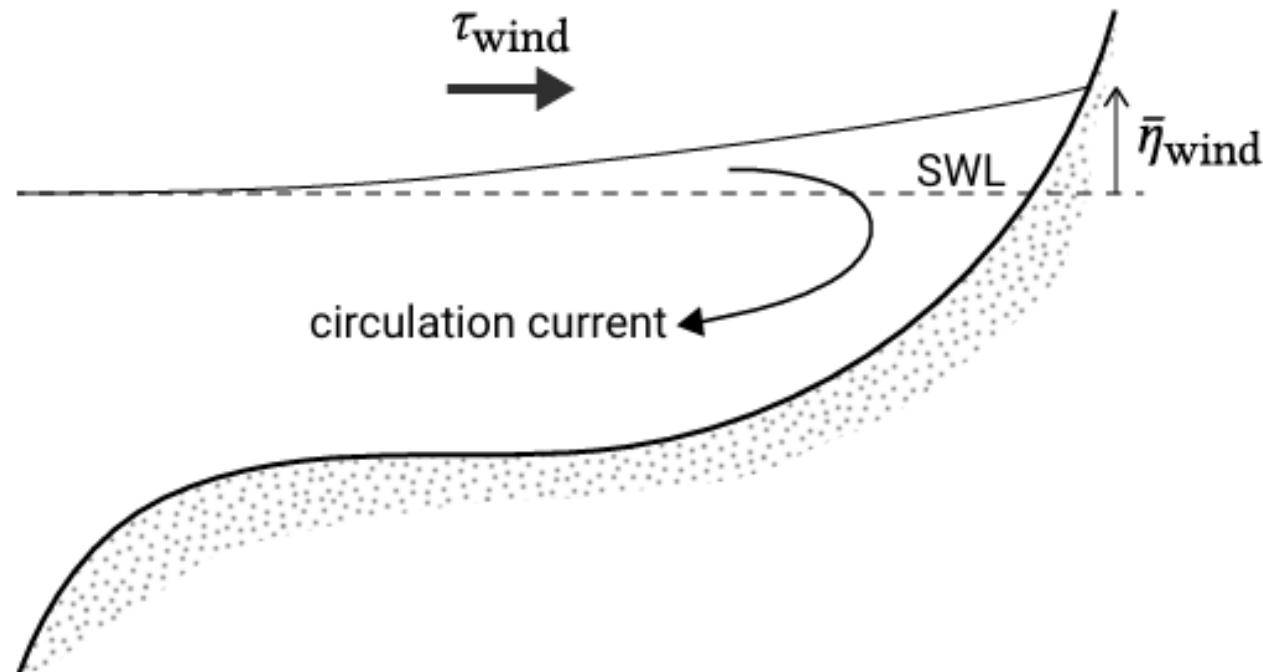
C_d = drag coefficient depending on wind velocity:

$C_d = (0.63 + 0.066W) \times 10^{-3}$ for $2 < W < 21$ (Smith & Banke, 1975)

ρ_a = density of air (1.25 kg/m³) in kg/m³

W = wind velocity at the water surface in m/s

- **Wind shear stress** causes the upper water layers to move in the **wind's direction**.
- A **seaward wind** creates a seaward current in the upper layers, while a **landward wind** creates a landward current.
- However, the coastline blocks the landward current, so in equilibrium, the net onshore flow must be zero.
- To balance this, **opposite-directed currents** form in the lower water layers. Additionally, a **water level set-up or set-down** occurs near the coast to counteract the wind-induced shear stresses.

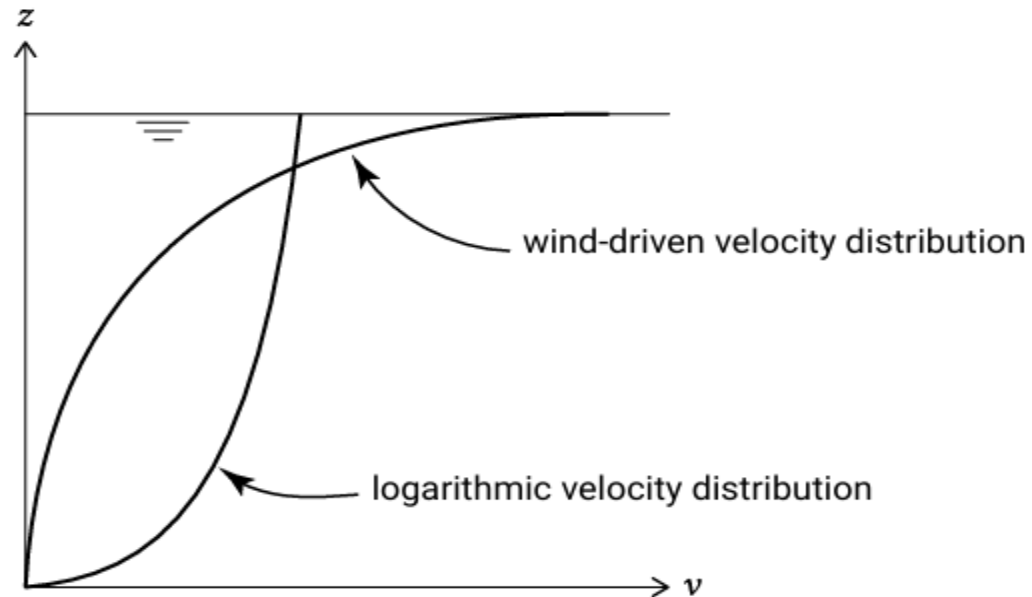


- The equilibrium condition is given by:

$$\rho g h \frac{d\bar{\eta}}{dx} = \tau_{\text{wind},x}$$

- Wind set-up is **inversely proportional to water depth**, causing water to pile up significantly in shallow coastal areas (e.g., storm surges).
- When wind **blows parallel to the coast**, it generates a **longshore current**, which takes about a day to fully develop.
- In equilibrium, without other forces, the **wind shear stress** balances the **bed shear stress**, similar to wave-induced currents.

- The vertical distribution of the **wind-generated current differs** substantially from the current generated by a water level gradient (see figure).



- The **highest flow velocities** occur at the water surface, with usually a **rapid decrease** in the downward direction (much more than with the **logarithmic velocity profile**).

- During storms, **wind stress** can significantly influence the **residual longshore current**.
- However, in the **littoral zone**, its effect is often negligible, and its **morphological impact** is limited due to low velocities near the seabed, where sediment concentrations are highest.
- Wind-driven currents are more important in areas like **coastal lagoons**.

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Problems

A steady wind of velocity 15 m/s blows over a shallow rectangular lagoon of uniform depth 5 m and length 20 km. The wind shear stress acting on the water surface follows the quadratic law Where drag coefficient $C_d = (0.63 + 0.066W) \times 10^{-3}$ and the air density is 1.25 kg/m³, steady one-dimensional flow and neglect bottom friction and Coriolis effects. Determine the wind shear stress?

Solution: drag coefficient

$$C_d = (0.63 + 0.066W) \times 10^{-3}$$

$$C_d = (0.63 + 0.066 \times 15) \times 10^{-3}$$

$$C_d = 1.62 \times 10^{-3}$$

$$\tau_{\text{wind}} = C_d \rho_a W^2$$

$$\tau_{\text{wind}} = 1.62 \times 10^{-3} \times 1.25 \times 15^2$$

$$\tau_{\text{wind}} = 0.456 \text{ N/m}^2$$